

Impact Statement: Agricultural Extension RAG Retrieval Benchmark

Summary

This benchmark trains and evaluates retrieval systems that match a smallholder farmer’s question to relevant agricultural-extension documents — the retrieval component of a RAG pipeline. Its downstream purpose is to make expert agronomic knowledge easier to surface for users who may not know the right search terms, the name of a disease, or which extension body to consult. The statement below considers who benefits, who is exposed to risk, and what should constrain deployment beyond the benchmark itself.

Intended Positive Impact

- **Lowens the retrieval barrier to expert knowledge.** Farmers rarely phrase problems the way agronomy literature does (“yellowing leaves” vs. “chlorosis, nitrogen deficiency”). A retriever tuned on this benchmark helps close that vocabulary gap.
- **Rewards intent-correct retrieval, not just keyword overlap.** The graded relevance scale and hard negatives (same crop/problem, wrong intent — e.g. prevention vs. causes) push systems toward the kind of disambiguation a real advisory tool needs, rather than superficial lexical matching that would surface confidently wrong advice.
- **Supports a broader RAG pipeline responsibly.** As a retrieval-only benchmark, it isolates and improves the step that determines *what an LLM is even allowed to read* before answering — better retrieval reduces the chance of the generation step hallucinating or answering from ungrounded prior knowledge.
- **Regional relevance.** Focused on Sub-Saharan African smallholder agriculture, a context that is underserved by mainstream agronomic tooling and datasets built around Global North farming systems.

Who Is Affected

Group	Nature of exposure
Smallholder farmers (end users, indirect)	Would receive retrieved documents (or downstream RAG answers) that inform real crop-management decisions — the group with the most to gain and the most to lose from retrieval errors
Model/RAG-system builders	Direct users of the benchmark; their retriever’s nDCG@5 score does not certify safety for deployment
Extension organizations (FAO, CGIAR, Plantwise, IITA, ICRISAT, AGRA, etc.)	Referenced stylistically or as licensors of grounded content; not evaluated or endorsed by benchmark performance

Risks and Potential Harms

- **Not agronomic advice, but easily mistaken for it.** The dataset explicitly states synthetic and LLM-rewritten text may contain simplifications or errors. A high nDCG@5 score reflects ranking quality against benchmark labels — it does not certify the retrieved *content* is agronomically correct. Any system built on this benchmark that is deployed toward real farmers needs a separate content-accuracy review, not just a retrieval-quality one.
- **False confidence from good leaderboard performance.** A strong score can create a false impression that the underlying knowledge base is production-ready. The benchmark measures retrieval behavior on a closed 695-document corpus, not the completeness or correctness of that corpus against real-world extension knowledge.
- **Consequences of wrong-intent retrieval are asymmetric.** Because it's agriculture, an error is not just "irrelevant search result" — a farmer acting on a prevention document when they needed a treatment document (or vice versa) can mean a lost season or wasted input costs. The hard-negative design mitigates this at the benchmark level but doesn't eliminate it in deployment.
- **Language and access gap.** The benchmark and corpus are English-only. The people it is nominally built to serve — smallholder farmers — are frequently more fluent in local languages. A system that scores well here may still be unusable by its intended beneficiaries without a translation or localization layer that this benchmark does not evaluate.
- **Corpus incompleteness treated as ground truth.** Relevance labels are only defined relative to the fixed 695-document corpus. A perfectly-scoring retriever can still fail a farmer whose real problem simply isn't represented in the corpus at all — the benchmark cannot distinguish "no relevant document retrieved" from "no relevant document exists."

Fairness and Representation Considerations

- **Geographic scope is explicit and narrow:** Sub-Saharan African smallholder agriculture. The card explicitly cautions against generalizing performance to other regions or farming systems — a retriever tuned here should not be assumed to transfer to, e.g., temperate large-scale agriculture.
- **Source mix:** a blend of synthetic (CC0, competition-authored) and `llm_grounded` (derived from CC-BY/CC0 sources such as CGIAR/CGSpace) documents. Because the synthetic half is not sourced from field-verified extension text, its accuracy and its coverage of locally-specific practices depend entirely on the quality of the generation process rather than on vetted agronomic institutions.
- **No demographic data on farmers is collected or required** — the benchmark operates on questions and documents only, which limits some fairness risks (no personal data) but also means nothing about *which* farmers' phrasing patterns are best or worst represented in the query set can be audited from the data alone.

Environmental / Resource Considerations

- Corpus and query volumes are small (695 documents, 308+200

queries), so both the TF-IDF baseline and most competitive retrievers can be trained and evaluated cheaply — this is a low-compute benchmark by design, which keeps participation accessible rather than favoring large-compute teams.

Recommendations for Responsible Use

1. **Treat this as a retrieval benchmark, not a deployment certification.** A leaderboard score answers “does this system find the labeled-relevant documents in the right order,” not “is this system safe to put in front of farmers.”
2. **Any real-world RAG system built on this corpus should add a content-verification step** before generation-stage answers reach an end user, given the corpus explicitly contains synthetic and LLM-rewritten passages.
3. **Do not extrapolate performance across geography or crop coverage** beyond what’s in the 695-document corpus.
4. **Plan for localization** (language, dialect, and channel — e.g. SMS/voice vs. text) before considering any deployment toward the stated beneficiary group; this benchmark evaluates none of that.
5. **Preserve attribution** for `llm_grounded` documents per their source license when reusing or redistributing corpus content.

Limitations of This Statement

This statement is derived entirely from the dataset documentation provided (competition README/data card). It does not reflect an independent audit of the corpus, the qrels’ annotation process, or the generation pipeline’s error rate — those remain open questions noted in the accompanying data card.